University of Florida
College of Public Health & Health Professions Syllabus
PHC 6064 Survey of Advanced Biostatistical Methods for the Health Sciences
3 credits, Spring 2026
Delivery Format: On-Campus Canvas: http://elearning.ufl.edu

Instructor:	Jonathan Fischer, PhD
Office:	5223 CTRB
Class meetings:	HPNP G111 (Tue 1:55 – 2:45 pm, Thu 1:55 – 3:50 pm)
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Email Address:	jfischer1@ufl.edu
Office Hours:	Mon and Tue 3:00 – 4:00 pm or by appointment In my office and/or on Zoom (link in Canvas)
Teaching Assistants:	TBD
Preferred Course Communications:	Canvas inbox

Prerequisites: *PHC6052 or permission of instructor. You should have at least one prior course in statistics and some exposure to R or SAS.*

PURPOSE AND OUTCOME

Course Overview
Uniquely blends the fundamentals of biostatistical inference with an introduction to advanced statistical techniques critical for the analysis of the growing compendium of health-related data. Topics span the analysis of high-dimensional, categorical, and longitudinal data from the health sciences. Applications utilize the statistical software packages R and SAS.

Course Description
A survey of biostatistical methods beyond basic testing and inference procedures. This course provides an overview of the advanced statistical procedures which are necessary for quantitative analysts and practitioners in the health sciences. Topics include complex regression models for independent, structured, and high-dimensional data, model/variable selection, and multiple testing, plus strategies for imputing missing data, multivariate parametric and non-parametric methods with applications to health-related data. Students will have the option to use either the R or SAS software packages.

Relation to Program Outcomes
This is a biostatistics course designed for students in public health and the health sciences and is also appropriate for students in the biological sciences looking to expand their quantitative analysis capabilities. It will train students to rigorously analyze potentially complicated data, enabling the assessment of public health programs as well as the critical evaluation and/or production of original research in clinical and academic settings. Moreover, students will develop their quantitative communication and presentation skills.

Course Objectives and/or Goals
Upon completion of the course, students will be able to:

1) Apply statistical methods to categorical, count, or structured data that commonly arise in the health and biological sciences;
2) Quantify the impact of missing data and employ suitable methods in response, particularly for the censoring and missingness patterns that frequently occur in clinical and biological data;

3) Analyze patient survival data in epidemiological contexts;
4) Implement advanced dimension reduction and visualization tools to deal with high dimensional data, such as those arising in genomics;
5) Employ simple machine learning approaches for classification and clustering with applications in personalized medicine and case/control studies; and
6) Develop their own testable hypotheses and effectively communicate biostatistical techniques and results to an audience of clinicians or health workers, both via written reports and oral presentations.

Instructional Methods
Class sessions consist of a mix of lectures and case studies with data using R and/or SAS. Data sets, code, and lecture notes are available to students in Canvas. Examples make exclusive use of data from public health, clinical, biomedical, and genomic settings, and features unique to data from these sources are emphasized. Assessments consist of weekly quizzes and homework assignments plus a course project. The course covers methods for independent, correlated, and multivariate data. Examples of typical questions we can answer with these methods are shown below.

Topical Outline/Course Schedule
Advanced Methods for Independent Data: What are the optimal tools when our data are independent but not amenable to the simplest approaches? For instance, how should we best predict disease status or incidence, and how can we explicitly control for confounding effects when assessing the effectiveness of treatments?
Methods for Correlated Data: How do we handle data which are correlated, whether hierarchically or temporally? Examples include comparing measurements when some patients share the same hospital or physician, monitoring metrics of public health over time, and tracking patient disease survival times.
Multivariate Data Methods: How can we group or classify individuals based on clinical or genomic covariates? For example, are there subtypes within a certain disease?

Days	Topic
Jan 13 & 15	Contingency tables
Jan 20 & 22	Multiple linear regression fundamentals
Jan 27 & 29	Model selection in linear regression
Feb 3 & 5	Linear regression diagnostics & extensions
Feb 10 & 12	GLMs – logistic regression
Feb 17 & 19	GLMS – count regression
Feb 24 & 26	Dimension reduction & clustering
Mar 3 & 5	Spatiotemporal data basics
Mar 10 & 12	Linear mixed models (fixed versus random effects)
Mar 24 & 26	Linear mixed models (model fitting and interpretation)
Mar 31 & Apr 2	Linear mixed models (inference)
Apr 7 & 9	GLMMs and GEEs
Apr 14 & 16	Missing data methods
Apr 21	Multiple hypothesis testing
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Course Materials and Technology
Texts: There are no required texts for this course. Below are several references.
- (RMB) Vittinghoff, E., Glidden, D.V., Shiboski, S.C., and C.E. McCulloch. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models.* Second Edition. Springer: New York, 2012.

- (AMS) Der, G. and B.S. Everitt. *Applied Medical Statistics Using SAS*. CRC Press: Boca Raton, 2013.
- (ISLA) James, G., Witten, D., Hastie, T., and R. Tibshirani. *An Introduction to Statistical Learning with Applications in R*. Springer: New York, 2013.

Software: Students will need to use either R or SAS. R and RStudio can be downloaded free from https://www.r-project.org/ and https://posit.co/. SAS can be accessed for free using SAS OnDemand for Academics (https://www.sas.com/en_us/software/on-demand-for-academics.html).

For technical support for this class, please contact the UF Help Desk at:

- Learning-support@ufl.edu
- (352) 392-HELP - select option 2
- https://lss.at.ufl.edu/help.shtml

ACADEMIC REQUIREMENTS AND GRADING

Assignments

- Twelve **homework assignments** will be assigned and evaluated for both completion and correctness. They will consist of guided statistical analyses using software and provided data sets with follow-up questions regarding the generated output. These will allow students to practice analyzing data using methods from class and to interpret and explain their results.
- Each week, students will complete a **quiz** to ensure they are comfortable with the core concepts from the lectures and readings. These short quizzes will generally consist of 5-10 multiple choice or multiple answer style questions.
- Each student will complete a **final project** individually with a data set of their choice using methods covered in the course. This process will include a proposal, mid-semester update, code and output, short written summary, and presentation. The summary should walk through the scientific background and motivation before discussing the analysis, its conclusions, and any implications. Projects will be assessed on the basis of accuracy and clarity, and additional guidance regarding these expectations, along with assistance selecting a suitable data set, will be provided during the course. TurnItIn will be used to evaluate reports for originality.

Grading

Assignment	Percent of final grade	Notes
Quizzes	15%	15 quizzes, 1% each
Homework	60%	12 assignments, 5% each
Project	25%	Proposal (5%) Status update (5%) Code/output (5%) Report (5%) Presentation (5%)

Please be aware that a C- is not an acceptable grade for graduate students. The GPA for graduate students must be 3.0 based on 5000 level courses and above to graduate. A grade of C counts toward

a graduate degree only if based on credits in courses numbered 5000 or higher that have been earned with a B+ or higher.

Percentage Earned	Letter Grade
[93,100]	A
[90,93)	A-
[87,90)	B+
[83-87)	B
[80-83)	B-
[77,80)	C+
[73,77)	C
[70,73)	C-
[67,70)	D+
[63,67)	D
[60,63)	D-
< 60	E

Letter Grade	Grade Points
A	4.0
A-	3.67
B+	3.33
B	3.0
B-	2.67
C+	2.33
C	2.0
C-	1.67
D+	1.33
D	1.0
D-	0.67
E	0.0
WF	0.0
I	0.0
NG	0.0
S-U	0.0

More information on UF grading policy may be found at https://catalog.ufl.edu/graduate/regulations/

Policy Related to Make up Exams or Other Work
Please notify me as soon as possible if unanticipated circumstances arise which interfere with your ability to complete an assignment on-time. Late work without prior notification and approval (unless there is an emergency) will receive a zero. Late work with prior approval (or due to an emergency) will be eligible for full credit. Any requests for make-ups due to technical issues must be accompanied by the ticket number received from e-learning support when the problem was reported to them. The ticket number will document the time and date of the problem. You must e-mail your instructor within 24 hours of the technical difficulty if you wish to request a make-up.

Policy Related to Required Class Attendance
We will follow the UF Attendance Policy. Please note all faculty are bound by the UF policy for excused absences. Excused absences must be consistent with university policies in the Graduate Catalog (https://catalog.ufl.edu/graduate/regulations/).

STUDENT EXPECTATIONS, ROLES, AND OPPORTUNITIES FOR INPUT

Expectations Regarding Course Behavior
Students are expected to show up for class prepared and on-time. Cell phones are to be silenced during class unless there is an emergency, in which case please inform the instructor.

Communication Guidelines
Please see the guidelines linked here: http://biostat.ufl.edu/current-students/e-learning-resources/e-learning-basics/etiquette-online/.

Academic Integrity

Students are expected to act in accordance with the University of Florida policy on academic integrity. As a student at the University of Florida, you have committed yourself to uphold the Honor Code, which includes the following pledge:

"**We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honesty and integrity**."

You are expected to exhibit behavior consistent with this commitment to the UF academic community, and on all work submitted for credit at the University of Florida, the following pledge is either required or implied:

"On my honor, I have neither given nor received unauthorized aid in doing this assignment."

It is your individual responsibility to know and comply with all university policies and procedures regarding academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action. For additional information regarding Academic Integrity, please see Student Conduct and Honor Code or the Graduate Student Website for additional details:
https://www.dso.ufl.edu/sccr/process/student-conduct-honor-code/
http://gradschool.ufl.edu/students/introduction.html

Please remember cheating, lying, misrepresentation, or plagiarism in any form is unacceptable and inexcusable behavior.

Online Faculty Course Evaluation Process
Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/.

Policy Related to Guests Attending Class:
Only registered students are permitted to attend class. However, we recognize that students who are caretakers may face occasional unexpected challenges creating attendance barriers. Therefore, by exception, a department chair or his or her designee (e.g., instructors) may grant a student permission to bring a guest(s) for a total of two class sessions per semester. This is two sessions total across all courses. No further extensions will be granted. Students are responsible for course material regardless of attendance. For additional information, please review the Classroom Guests of Students policy in its entirety. Link to full policy: http://facstaff.phhp.ufl.edu/services/resourceguide/getstarted.htm

SUPPORT SERVICES

Accommodations for Students with Disabilities
Students with disabilities who experience learning barriers and would like to request academic accommodations should connect with the disability Resource Center. Click here to get started with the Disability Resource Center. It is important for students to share their accommodation letter with their instructor and discuss their access needs, as early as possible in the semester.

Counseling and Student Health
Students sometimes experience stress from academic expectations and/or personal and interpersonal issues that may interfere with their academic performance. If you find yourself facing issues that have the potential to or are already negatively affecting your coursework, you are encouraged to talk with an instructor and/or seek help through University resources available to you.

- The Counseling and Wellness Center 352-392-1575 offers a variety of support services such as psychological assessment and intervention and assistance for math and test anxiety. Visit their web site for more information: http://www.counseling.ufl.edu. On line and in person assistance is available.

- You Matter We Care website: http://www.umatter.ufl.edu/. If you are feeling overwhelmed or stressed, you can reach out for help through the You Matter We Care website, which is staffed by Dean of Students and Counseling Center personnel.

- The Student Health Care Center at Shands is a satellite clinic of the main Student Health Care Center located on Fletcher Drive on campus. Student Health at Shands offers a variety of clinical services. The clinic is located on the second floor of the Dental Tower in the Health Science Center. For more information, contact the clinic at 392-0627 or check out the web site at: https://shcc.ufl.edu/

- Crisis intervention is always available 24/7 from:
  Alachua County Crisis Center: (352) 264-6789
  http://www.alachuacounty.us/DEPTS/CSS/CRISISCENTER/Pages/CrisisCenter.aspx

Do not wait until you reach a crisis to come in and talk with us. We have helped many students through stressful situations impacting their academic performance. You are not alone so do not be afraid to ask for assistance.

Inclusive Learning Environment
Public health and health professions are based on the belief in human dignity and on respect for the individual. As we share our personal beliefs inside or outside of the classroom, it is always with the understanding that we value and respect diversity of background, experience, and opinion, where every individual feels valued. We believe in, and promote, openness and tolerance of differences in ethnicity and culture, and we respect differing personal, spiritual, religious and political values. We further believe that celebrating such diversity enriches the quality of the educational experiences we provide our students and enhances our own personal and professional relationships. We embrace The University of Florida's Non-Discrimination Policy, which reads, "The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act." If you have questions or concerns about your rights and responsibilities for inclusive learning environment, please see your instructor or refer to the Office of Multicultural & Diversity Affairs website: www.multicultural.ufl.edu